

Communication

Skin deformation analysis for pre-operative planning of DIEAP flap reconstruction surgery

Rafaela Timóteo^{a,b,*}, David Pinto^b, Marta Martinho^b, Pedro Gouveia^{b,d},
Daniel Simões Lopes^{a,c,e}, Carlos Mavioso^b, Maria João Cardoso^b

^a Instituto Superior Técnico, Universidade de Lisboa, Av. Rovisco Pais, 1049-001 Lisboa, Portugal

^b Breast Unit/Digital Surgery Lab, Champalimaud Clinical Centre/Champalimaud Foundation, Avenida Brasília, 1400-038 Lisboa, Portugal

^c INESC ID, Rua Alves Redol 9, 1000-029 Lisboa, Portugal

^d Faculdade de Medicina de Lisboa, Av. Prof. Egas Moniz MB, 1649-028 Lisboa, Portugal

^e ITI/LARSys, Hub Criativo do Beato, Factory Lisbon, Rua da Manutenção 71, Building F S05, 1900-500 Lisboa, Portugal

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ABSTRACT

Deep inferior epigastric artery perforator (DIEAP) flap reconstruction surgeries can potentially benefit from augmented reality (AR) in the context of surgery planning and outcomes improvement. Although three-dimensional (3D) models help visualize and map the perforators, the anchorage of the models to the patient's body during surgery does not consider eventual skin deformation from the moment of computed tomography angiography (CTA) data acquisition until the position of the patient while in surgery. In this work, we compared the 3D deformation registration from supine arms down (CTA position) to supine with arms at 90° degrees (surgical position), estimating the patient's skin deformation. We processed the data sets of 20 volunteers with a 3D rigid registration tool and performed a descriptive statistical analysis and statistical inference. With 2.45 mm of root mean square and 2.89 mm of standard deviation, results include 30% cases of deformation above 3 mm and 15% above 4 mm. Pose transformation deformation indicates that 3D surface data from the CTA scan position differs from data acquired in loco at the surgical table. Such results indicate that research should be conducted to construct accurate 3D models using CTA data to display on the patient, while considering projection errors when using AR technology.

1. Introduction

DIEAP flap reconstruction surgery is a common technique for immediate or delayed breast reconstruction surgery for patients previously submitted for a mastectomy [1,2]. The DIEAP flap reconstruction consists of harvesting excess skin and/or subcutaneous fat from the abdomen for breast reconstruction [3]. The flap is vascularized through perforator vessels that are dissected and preserved along their trajectory until they reach the inferior epigastric vessels. The correct identification, preservation, and selection of the best perforator is a key step to proceeding with the breast's reconstruction. The perforators are manually identified, characterized, and mapped by a radiologist, commonly through CTA, allowing the surgeon to choose the most suitable for the

procedure [4-5].

During surgery, the perforators are marked on the patient's skin using the radiologist map and taking the umbilicus as a reference. The manual process is expensive, tedious, and has errors that are human-dependent. Our group developed a semiautomatic solution to track the vessels automatically and create an immediate report presented to the surgeon, in high agreement with the manual process [6]. The problem stands in accurately translating the perforators' location extracted from the CTA images to the patient's body in surgery, minimizing errors from changes in body positioning. These errors are often identified during the doppler identification and flap harvesting.

Recently, AR has been considered to assist surgeons during DIEAP reconstruction. As an interaction paradigm that merges virtual objects in

* Corresponding author.

E-mail addresses: rafaela.timoteo@research.fchampalimaud.org (R. Timóteo), david.pinto@fundacaochampalimaud.pt (D. Pinto), marta.martinho@fundacaochampalimaud.pt (M. Martinho), pedro.gouveia@fundacaochampalimaud.pt (P. Gouveia), daniel.s.lopes@tecnico.ulisboa.pt (D.S. Lopes), carlos.mavioso@fundacaochampalimaud.pt (C. Mavioso), maria.joao.cardoso@fundacaochampalimaud.pt (M.J. Cardoso).

the real-world environment, the Microsoft HoloLens 2¹ is a head-mounted AR device that can potentially enable breast surgeons to work hands-free while having access to relevant patient-specific information in an augmented space [7-11]. Besides the HoloLens, previous research has contemplated augmentations of CTA-based perforator data through other head-mounted displays (HMD), projectors, and digital tablets, obtaining registration errors in the range of 1.7 to 8.8 ± 6.6 mm [7,10-15]. However, the HoloLens presents several benefits such as: (i) hands-free task performance; (ii) voice and hand gesture recognition for interacting with 3D content; (iii) eye-tracking; (iv) 3D visualization of data blended with reality; and (v) robust battery life [16]. The HoloLens have been predominantly used for medical purposes and research, including during breast cancer surgery to visualize the tumor inside the patient's breast [17-19].

Nonetheless, previous research approaches were not validated, dismissing potential deformation of the patient's lower abdomen due to different image registration positions between the CTA and operating table positions. A study to ascertain the amount of this type of deformation is still lacking, thus all works rely on the assumption that the deformation is irrelevant when projecting the models on patients. This work's purpose is to evaluate the degree of deformation occurring when performing the fusion of the CTA scan with the surface scan of the patient on the operating table.

2. Materials and methods

Twenty female volunteers from our Breast Unit staff were invited and consented to image acquisition sessions with 3D surface scanning in two different positions using the Go!Scan 3D handheld (CreaformTM²). The volunteers' mean body mass index (BMI) was 24,3 (range of 20,5 - 30,9). The volunteer's first position corresponded to the patient's position during the CTA Scan: arms down and close to the body. The second position corresponded to the patient's position during surgery: arms open at 90° (Fig. 1(a) and (b)).

The 3D surface scans contain irrelevant points that can negatively affect the results if considered during mesh registration (Fig. 1(c) and (d)). Since the skin area at the abdominal and lower abdominal levels are our focus, every data point outside this area was removed using the MeshLab³ software by manually selecting rectangular areas.

Afterward, we used CloudCompare⁴ software for mesh alignment and statistical analysis. We used different alignment methods to ensure that the deformation values were not affected by how we performed the alignment. Each pair of meshes was aligned using seven different combinations of three distinct types of alignment for a more precise evaluation of the deformation: Iterative Closest Point (ICP), manual alignment, and alignment with point pairs for a more precise evaluation of the deformation (Fig. 2).

Three of the seven mesh alignment combinations were selected to align each pair of meshes:

1. Iterative Closest Point (ICP) Algorithm [20]: automatic alignment for rigid registration, employed to minimize the difference between two similar and pre-registered point clouds by iterating between the closest point query, converging to an optimal alignment.
2. Manual and ICP (MI): manual alignment using the touchpad to control the position and rotation of the meshes until they are registered roughly, followed by the performance of the ICP algorithm.
3. Six (6) landmarks and ICP (LI): alignment by registration of six landmarks on the abdomen, registered approximately on the same

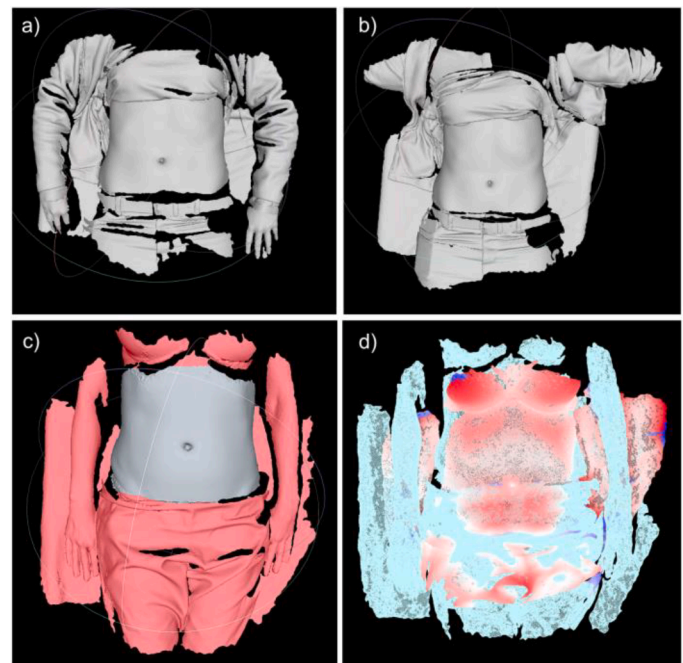


Fig. 1. (a) Volunteer in the patient's position during the CTA scan and (b) the patient's position during surgery. (c) Surface scan including the capture of irrelevant parts of the body (in red) leading to incorrect alignment of the meshes; and (d) resulting in a majority of red points on the color map after alignment.

locations in each mesh using the umbilicus as reference. Meshes were aligned according to the landmarks, and then the ICP algorithm was applied for better adjustment (or fine-tuning) (Fig. 3).

Once the meshes were aligned, we proceeded to compute the distance between the point clouds automatically using the Cloud-to-Mesh Distance (C2M) algorithm. Each point of one cloud searched for the nearest triangle in the other cloud during computation. After the computation, the compared cloud revealed a color scale representing the distance between the points from it to the reference cloud (Fig. 3(c)) and three outputs were returned: mean distance, standard deviation, and root mean square (RMS).

3. Results

Skin deformation analysis performed in this work indicates that the best results were achieved using ICP (mean distance of 0.03 mm and RMS of 2.42 mm), the combination of manual alignment plus ICP (mean distance of 0.04 mm and RMS of 2.363 mm), and the alignment using six landmarks plus ICP (mean distance of 0.11 mm and RMS of 2.57 mm).

For each volunteer, we find slight color changes in the rows of the heat map presented in (Table 1). However, there were some offset values, such as the results obtained from the C2 volunteer: by using ICP and MI methods of alignment it was obtained the RMS of 2.34 mm and 2.35 mm, respectively. We noted that when using the LI method the value escalated to 4.54 mm. It is also possible to observe three cases of deformation in which the values of RMS were above 4 mm (C1, C6, C18) and three cases in which the values of RMS were above 3 mm (C3, C8, C16), corresponding to 15% and 30% of the results, respectively.

The average of the RMS obtained, using the three methods described, is 2.45 mm (range of 0.65 - 5.05 mm). The standard deviation calculated is 2.89 mm. An histogram containing the sum of all points from all volunteers' point clouds (meshes) and their distance from the reference mesh - after manual alignment followed by the performance of the ICP algorithm - shows a majority of points clustered between the interval of

¹ <https://www.microsoft.com/en-us/hololens>

² <https://www.creaform3d.com/en/handheld-portable-3d-scanner-goscan-3>

³ <https://www.meshlab.net/#>

⁴ <https://www.cloudcompare.org/main.html>

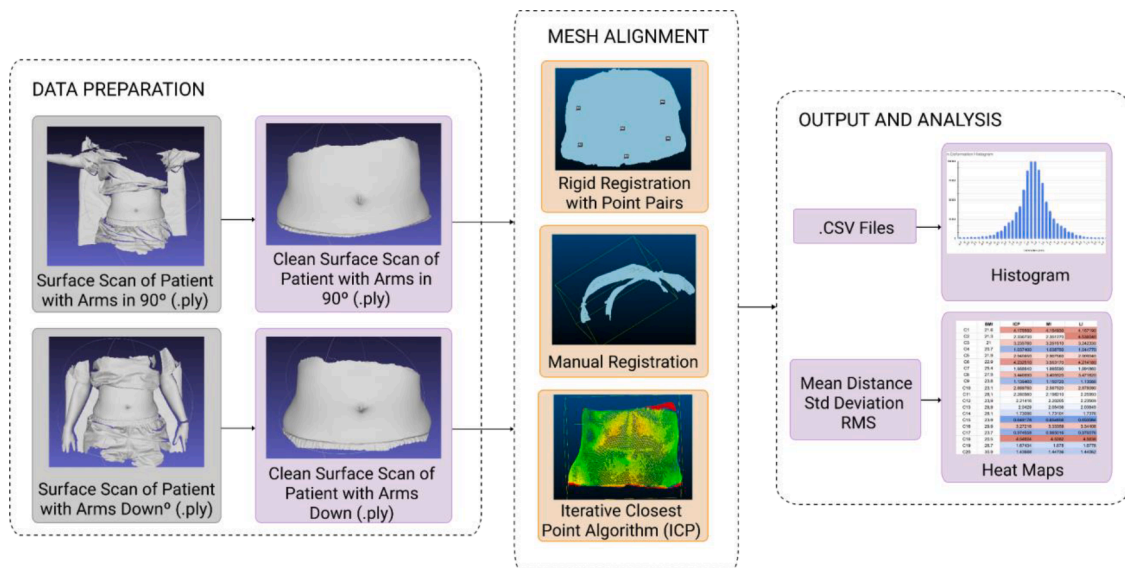


Fig. 2. Skin deformation analysis pipeline. The acquired meshes were cleaned using MeshLab by removing irrelevant data points such as arms, scanned dust, or clothing. Each cleaned pair of meshes was imported into CloudCompare and aligned using mesh alignment techniques. Afterward, four statistical outputs were provided for deformation analysis: histogram, mean distance, standard deviation, and root mean square.

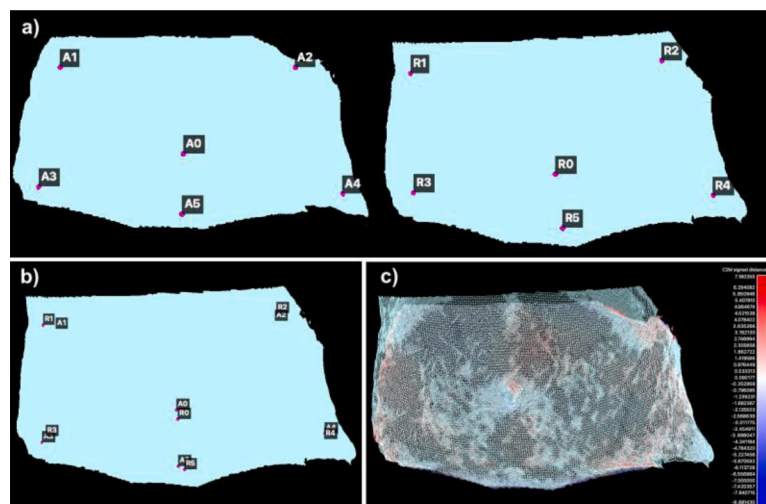


Fig. 3. (a) Six landmarks plus ICP algorithm: six points were registered on the two meshes. (b) These points are used as references for the alignment, which was then followed by the ICP algorithm. (c) Color scale of the distances between the meshes after the alignment was obtained.

[-5; 5] mm.

4. Discussion

When projecting 3D models of registered angio- CT perforators during free flap DIEAP reconstruction, previous research did not account for errors resulting from deformation of the patient's abdominal wall during data acquisition (Table 2). These studies highlight registration errors that challenge the conventional implementation of such technologies in high-precision procedures. To create an accurate 3D model for breast cancer surgery, Gouveia et al. created a digital breast with a tumor from a fusion between a breast MRI with pose transformation and a 3D surface scan of the patient's torso and evaluated the 3D volume estimation of the breast and the accuracy of the breast surface scan with the breast MRI [21].

To the best of our knowledge, this is the first paper to perform descriptive statistical analysis and statistical inference for two different registration positions (closed arms versus open arms) and quantify

target registration errors between the two positions. For each subject, it was confirmed that the different alignment methods produced consistent results.

Deformation errors can occur during data acquisition, 3D model reconstruction and alignment of the mesh on the patient during surgery [22]. Deformation of the skin mesh can affect the construction of the 3D model and result in an inaccurate display of the 3D model when it is anchored to the patient's body. A high-fidelity projection of the 3D model requires error control at all stages, which consist of the use of different instruments. In this paper, we analyze the potential error at the beginning of the pipeline for the construction and display of the 3D model: data acquisition. It is not possible to recreate the position of the patient in the OR during a CTA acquisition, mainly for technical reasons: The scanner tunnel does not provide enough space for an open position of the arms, as the arms have to be placed close to the body or above the head - which would lead to a larger deformation of the breast. Therefore, the deformation of the breast must be quantified.

The results disclose that the deformation error caused by the

Table 1

Root Mean Square heat map of the results obtained from the 20 volunteers' meshes using the three outstanding methods of alignment: iterative closest point (ICP); manual with ICP (MI); six landmarks with ICP (LI).

	BMI	ICP	MI	LI
C1	21.6	4.175550	4.154930	4.157190
C2	21.3	2.336730	2.351270	4.536040
C3	21	3.235780	3.251510	3.242330
C4	20.7	1.037400	1.038750	1.044770
C5	21.9	2.945650	2.907560	2.909040
C6	22.9	4.232510	3.553170	4.214180
C7	25.4	1.988640	1.995590	1.991860
C8	27.9	3.446690	3.405920	3.471820
C9	23.8	1.139460	1.150720	1.13086
C10	23.1	2.889780	2.587520	2.878090
C11	28.1	2.280980	2.198210	2.25993
C12	23.9	2.21416	2.20205	2.23569
C13	29.9	2.0429	2.05436	2.03845
C14	28.1	1.73096	1.73101	1.7376
C15	23.9	0.649178	0.654856	0.650084
C16	29.9	3.27216	3.33359	3.34406
C17	23.7	0.974558	0.985016	0.976076
C18	20.5	4.54924	4.5282	4.5636
C19	26.7	1.67434	1.678	1.6776
C20	30.9	1.43688	1.44736	1.44062

Table 2

Previous studies with simple holographic projection on DIEAP flap reconstruction surgeries.

Author	Year of Publication	Number of Patients	Visualization Method	Registration Error
Hummelink et al. [13]	2018	60	Projection (2D)	Not Mentioned
Bosc et al. [12]	2017	12	Stereoscopic smart glasses	Not Mentioned
Wesselius et al. [8]	2020	1	HoloLens™	8.8 ± 6.6 mm
Jiang et al. [10]	2017	1 (an animal)	HMD using ARToolKit software	3.47 ± 1.55 mm
Gonzalez et al. [14]	2021	5 (simulations with mannequins)	VascuLens projector-based AR system	From 1.7 to 4.3 mm
Fitoussi et al. [15]	2021	12	Digital Tablet (AR-assisted)	2 mm
Cercenelli et al. [11]	2022	2 (simulations with a mannequin)	Digital Tablet and HoloLens 2	1 to 5 mm

different positions of data acquisition must be taken into account when creating the models in order to more accurately determine the position of the perforators during surgery. It is also necessary to consider the errors obtained in the anchoring process using the Microsoft HoloLens, as identified in previous studies [7,11]. Depth perception and drift can cause the model to appear to deviate from its original position, compromising the accuracy of the model's registration [23].

Nonetheless, the value of deformation can be influenced by several factors: (i) the subject's breathing cycles can influence the capture of the mesh during surface scan acquisition, hence, mesh distortion can influence the subsequent alignment; and (ii) alignment methods might not result in error-free alignment of the mesh pairs, leading to particular cases of offsetting values that should not be a concern despite their influence on the final results.

Since there are no well-defined anatomical structures in the abdominal region (besides the umbilicus) for registration, we opted for a markerless approach. Skin marks, such as the nevus, are present in most patients, but may not be in sufficient quantity. Another option would be to mark specific points (e.g. with a ruler and pen) for a marker approach (i.e. with ArUco markers). However, such an option entails several limitations: (i) increased time for the preoperative moment; (ii) user-

caused errors when re-placing the markers in the OR; and (iii) the markers would take up space in the abdominal area where the perforators are marked.

The current study has the following limitations: (i) although the limited number of participants showed interesting and consistent results, a larger patient population could dictate more accurate statistics; (ii) further studies on the fusion of the CTA imaging transformed into a 3D model with the surface scans are needed: some deformations were not validated in this study, caused by curvatures of the tables in which the patient lies during data acquisition, especially during the CTA scan and later in the operating room; (iii) the BMI values used to find a relationship between the results obtained were inconclusive; the range of BMI values is small [20.5; 30.9] and future work should include a larger sample of patients with a wider range of values to investigate the correlations between the IBM values and skin deformity; and (iv) the landmarks used in the alignment methods could be previously marked on the surface scans and CTA scans for more accurate alignment. According to the distance to the umbilicus, these landmarks were selected on CloudCompare on both meshes, which showed inaccurate alignment without further adjustments (such as ICP).

5. Conclusion

This work confirmed that the deformation of the patient's skin from CTA data acquisition and operating table is relevant since it reaches an RMS of 2,45 mm with 30% of the cases above 3 mm and 15% of the cases above 4 mm. Thus, the deformation detected should be of concern. It is important to perform further research considering the errors in the projection of the final model in the operating table using the Microsoft HoloLens. In addition, more volunteers should be considered to also study the deformation's correlation with BMI values.

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Ethical approval

Not required.

Declaration of Competing Interest

None declared.

References

- [1] Seidenstuecker K, van Waes C, Munder BI, Claes KE, Witzel C, Roche N, Stillaert F, Mahajan AL, Andree C, Blondeel PN. DIEAP flap for safe definitive autologous breast reconstruction. *Breast* 2016;26:59–66. <https://doi.org/10.1016/j.breast.2015.12.005>.
- [2] Healy C, Allen Sr RJ. The evolution of perforator flap breast reconstruction: twenty years after the first DIEP flap. *J Reconstr Microsurg* 2014;30(02):121–6.
- [3] Munder B, Andree C, Witzel C, Fertsch S, Stambera P, Schulz T, Wolter A. The DIEP Flap as Well-established Method of Choice for Autologous Breast Reconstruction with a Low Complication Rate—Retrospective Single-centre 10-Year Experience. *Geburtshilfe Frauenheilkd* 2020;80(06):628–38.
- [4] Araújo RJ, Garrido V, Barac CA, Vasconcelos MA, Mavioso C, Anacleto JC, Oliveira HP. Computer aided detection of deep inferior epigastric perforators in computed tomography angiography scans. *Comput Med Imaging Graph* 2019;77: 101648.
- [5] Uppal RS, Casar B, Van Landuyt K, Blondeel P. The efficacy of preoperative mapping of perforators in reducing operative times and complications in perforator flap breast reconstruction. *J Plast Reconstr Aesthet Surg JPRAS* 2009;62(7):859–64. <https://doi.org/10.1016/j.bjps.2008.04.015>.
- [6] Mavioso C, Araújo RJ, Oliveira HP, Anacleto JC, Vasconcelos MA, Pinto D, Gouveia PF, Alves C, Cardoso F, Cardoso JS, et al. Automatic detection of perforators for microsurgical reconstruction. *The Breast* 2020;50:19–24.
- [7] Wesselius TS, Meulstee JW, Luijten G, Xi T, Maal TJJ, Ulrich DJO. Holographic Augmented Reality for DIEAP Flap Harvest. *Plast Reconstr Surg* 2021;147(1): 25e–9e. <https://doi.org/10.1097/PRS.00000000000007457>.

- [8] Salehahmadi F, Hajialiasgari F. Grand Adventure of Augmented Reality in Landscape of Surgery. *World J Plast Surg* 2019;8(2):135–45. <https://doi.org/10.29252/wjps.8.2.135>.
- [9] Birlo M, Edwards PE, Clarkson M, Stoyanov D. Utility of optical see-through head mounted displays in augmented reality-assisted surgery: a systematic review. *Med Image Anal* 2022;102361.
- [10] Jiang T, Zhu M, Zan T, Gu B, Li Q. A novel augmented reality-based navigation system in perforator flap transplantation—A feasibility study. *Ann Plast Surg* 2017;79(2):192–6.
- [11] Cerenelli L, Babini F, Badiali G, Battaglia S, Tarsitano A, Marchetti C, Marcelli E. Augmented reality to assist skin paddle harvesting in osteomyocutaneous fibular flap reconstructive surgery: a pilot evaluation on a 3D-printed leg phantom. *Front Oncol* 2022;11:5572.
- [12] Bosc R, Fitoussi A, Pigneur F, Tacher V, Hersant B, Meningaud JP. Repérage des vaisseaux perforants par réalité augmentée: application au lambeau perforant d'artère épigastrique inférieure profonde [Identification of perforating vessels by augmented reality: Application for the deep inferior epigastric perforator flap]. *Ann Chir Plast Esthet* 2017;62(4):336–9. <https://doi.org/10.1016/j.anplas.2017.01.002>.
- [13] Hummelink Stefan, Hoogveen Yvonne L, Kool Leo J, Schultze. Dietmar J O Ulrich A New and Innovative Method of Preoperatively Planning and Projecting Vascular Anatomy in DIEAP Flap Breast Reconstruction: A Randomized Controlled Trial. *Plast Reconstr Surg* 2019;143(6):1151e–8e. <https://doi.org/10.1097/PRS.0000000000005614>.
- [14] Gonzalez S, Stein M, Rohling R, Edgcombe P. The VascuLens: A Handsfree Projector-Based Augmented Reality System for Surgical Guidance During DIEP Flap Harvest. *CMBES Proc* 2021;44(May 2021).
- [15] Fitoussi A, Tacher V, Pigneur F, Heranney J, Sawan D, Dao TH, Bosc R. Augmented reality-assisted deep inferior epigastric artery perforator flap harvesting. *J Plast Reconstr Aesthet Surg* 2021;74(8):1931–71.
- [16] Tepper OM, Rudy HL, Lefkowitz A, Weimer KA, Marks SM, Stern CS, Garfein ES. Mixed reality with HoloLens: where virtual reality meets augmented reality in the operating room. *Plast Reconstr Surg* 2017;140(5):1066–70.
- [17] Gouveia PF, Costa J, Morgado P, Kates R, Pinto D, Mavioso C, Cardoso MJ. Breast cancer surgery with augmented reality. *The Breast* 2021;56:14–7.
- [18] Park S, Bokijonov S, Choi Y. Review of microsoft hololens applications over the past five years. *Appl Sci* 2021;11(16):7259.
- [19] Hanna MG, Ahmed I, Nine J, Prajapati S, Pantanowitz L. Augmented reality technology using Microsoft HoloLens in anatomic pathology. *Arch Pathol Lab Med* 2018;142(5):638–44.
- [20] He Ying, Liang Bin, Yang Jun, Li Shunzhi, He Jin. An Iterative Closest Points Algorithm for Registration of 3D Laser Scanner Point Clouds with Geometric Features. *Sensors* 2017;17:1862. <https://doi.org/10.3390/s17081862>.
- [21] Gouveia PF, Oliveira HP, Monteiro JP, Teixeira JF, Silva NL, Pinto D, Cardoso MJ. 3D breast volume estimation. *Eur Surg Res* 2021;62(3):121–6.
- [22] Bauer, M. (2007). Tracking Errors in Augmented Reality. PhD thesis.
- [23] Vassallo R, Rankin A, Chen EC, Peters TM. Hologram stability evaluation for Microsoft HoloLens. *Medical imaging2017: image perception, observer performance, and technology assessment* 2017;10136:295–300.